

The empirical recurrence for OEIS A233024: recovering a conjecture that is not written down

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2 September 2026

Abstract

OEIS A233024 records “Empirical recurrence of order 50”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 58$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 50, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 50 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

2020 Mathematics Subject Classification. 05A15, 11B37, 15A18.

1 A conjecture one cannot read

OEIS A233024 is “Number of $n \times 6$ 0..3 arrays with no element $x(i,j)$ adjacent to value $3-x(i,j)$ horizontally, vertically, diagonally or antidiagonally, and top left element zero.”. It has offset 1 and begins

243, 12555, 959385, 75489671, 6050598339, 487651369463, 39386531523609, 3183658674622011, ...

The entry states:

Empirical recurrence of order 50 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 07:47:36, revision 7) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 58$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 58$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S' = 116$ exact terms of the model returns order 50 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{50} c_i a(n-i),$$

c_1	137	c_{14}	11586072411107	c_{27}	15961099145793044249	c_{40}	−53852018579742548
c_2	−5154	c_{15}	2843616129254272	c_{28}	10040367024765624333	c_{41}	−41872056963315104
c_3	36794	c_{16}	428336461272989	c_{29}	−20284422829281519925	c_{42}	9919276041671256
c_4	1282850	c_{17}	−30578136816100160	c_{30}	−10523152782115952180	c_{43}	2078982585671984
c_5	−18244275	c_{18}	−11466765415259710	c_{31}	18783945076793311526	c_{44}	−728710831536352
c_6	−112255480	c_{19}	221059534574721721	c_{32}	7137462742733912937	c_{45}	−25930737591584
c_7	2514327616	c_{20}	121553605295182283	c_{33}	−12550621761898463641	c_{46}	22049402876032
c_8	4025796362	c_{21}	−1092304802266958969	c_{34}	−2901381906922957362	c_{47}	−620903148672
c_9	−173737235320	c_{22}	−721589424398039784	c_{35}	5904346076199200406	c_{48}	−240018990592
c_{10}	−35435131418	c_{23}	3753969916172607125	c_{36}	523968291233454310	c_{49}	7795918848
c_{11}	6983607062665	c_{24}	2657034679682609023	c_{37}	−1874338189139452042	c_{50}	945684480
c_{12}	−986354769526	c_{25}	−9129603433723281690	c_{38}	69653038613278764		
c_{13}	−175062224465997	c_{26}	−634830863844250417	c_{39}	374647021025747418		

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 50, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $50 + 4$ terms removed returns order 50 as well, so $\deg q_{\min} = 50 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 12 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $50 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 50 that this sequence satisfies — and that family turns out to have exactly one member.

References

- [1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org>, 2026. Sequence A233024.
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